To calculate the SAM and OAM continuity equations in an inhomogeneous medium, we adopt the Coulomb field $\mathbf{E}_{\text{Coulomb}} = -\nabla\varphi$ as the longitudinal electric field, where $\nabla^2\varphi = -\rho_P/\varepsilon_0$, and ρ_P is the polarized charge. Similarly, the inhomogeneous magnetization leads to a nonzero polarized magnetic charge, we adopt "Coulomb magnetic field" $\mathbf{H}_{\text{Coulomb}} = -\nabla\phi$ as the longitudinal magnetic field, where $\nabla^2\phi = -\rho_M$, and ρ_M is the is the polarized magnetic charge. Then, the total field becomes $\mathbf{E} = -\partial_t\mathbf{A} - \nabla\varphi$ and $\mathbf{H} = -\partial_t\mathbf{C} - \nabla\phi$. Under the Coulomb gauge ($\nabla\cdot\mathbf{A} = 0,\ \nabla\cdot\mathbf{C} = 0$), we have

$$\mathbf{S} = \frac{1}{2}(\varepsilon_0\mathbf{E}\times\mathbf{A} + \mu_0\mathbf{H}\times\mathbf{C} - \mathbf{P}\times\mathbf{A} + \mu_0\mathbf{M}\times\mathbf{C}),$$

$$\overleftrightarrow{\Sigma} = \frac{1}{2}\Big[\big(\mathbf{A}\cdot\mathbf{H} - \mathbf{A}\cdot\nabla\phi - \mathbf{C}\cdot\mathbf{E} + \mathbf{C}\cdot\nabla\varphi\big)\boldsymbol{I} - \mathbf{A}\otimes\mathbf{H} + \nabla\phi\otimes\mathbf{A} + \mathbf{C}\otimes\mathbf{E} - \nabla\varphi\otimes\mathbf{C}\Big],$$

$$\tau^{\text{spin}} = 2\,\partial_t(\mathbf{P}\times\mathbf{A}) + 2\,\mathbf{P}\times\mathbf{E} + 2\mu_0\,\mathbf{M}\times\mathbf{H},$$

$$\mathbf{L} = \frac{1}{2}\Big[\varepsilon_0\,(\mathbf{r}\times\nabla\mathbf{A})\cdot\mathbf{E} + \mu_0\,(\mathbf{r}\times\nabla\mathbf{C})\cdot\mathbf{H} - (\mathbf{r}\times\nabla\mathbf{A})\cdot\mathbf{P} + \mu_0\,(\mathbf{r}\times\nabla\mathbf{C})\cdot\mathbf{M}\Big],$$

$$\overleftrightarrow{\Lambda} = \frac{1}{2}\Big[(\mathbf{r}\times\nabla\mathbf{A})\times\mathbf{H} - (\mathbf{r}\times\nabla\nabla\phi)\times\mathbf{A} - (\mathbf{r}\times\nabla\mathbf{C})\times\mathbf{E} + (\mathbf{r}\times\nabla\nabla\varphi)\times\mathbf{C}\Big]^T,$$

$$\tau^{\text{orbit}} = \frac{1}{2}\Big\{2\,\partial_t\big[(\mathbf{r}\times\nabla\mathbf{A})\cdot\mathbf{P}\big] + (\mathbf{r}\times\nabla\mathbf{E})\cdot\mathbf{P} - (\mathbf{r}\times\nabla\mathbf{P})\cdot\mathbf{E}\Big\} + \frac{1}{2}\Big\{\mu_0\,(\mathbf{r}\times\nabla\mathbf{H})\cdot\mathbf{M} - \mu_0\,(\mathbf{r}\times\nabla\mathbf{M})\cdot\mathbf{H}\Big\},$$

$$\Delta^{\text{conversion}} = \frac{1}{2}\nabla\cdot\big(\mathbf{H}\otimes\mathbf{A} - \mathbf{A}\otimes\nabla\phi - \mathbf{E}\otimes\mathbf{C} + \mathbf{C}\otimes\nabla\varphi\big) + \nabla^2\phi\,\mathbf{A} - \nabla^2\varphi\,\mathbf{C},$$

where $\mathbf{A}$ is the magnetic vector-potential, and $\mathbf{C}$ is the electric vector-potential.

To analyze the interaction between light and NS, we first investigate the spin torque τ^{spin} inside the NS. Within the scope of Mie scattering, NS can no longer be regarded as an ideal dipole, because its size is comparable to the wavelength of light that the geometric dependency of the polarization P cannot be ignored. The interaction between the local field and local polarizability $\mathbf{P}$ leads to a local torque on the NS. Because $\mu_0\mathbf{M}\times\mathbf{H} = 0$ when $\mu_r = 1$, the spin torque τ^{spin} is dominated by the interaction between magnetic vector-potential $\mathbf{A}$ and current $\partial_t\mathbf{P}$, as $\partial_t\mathbf{P}\times\mathbf{A}$, together with the interaction between the transverse electric field $-\partial_t\mathbf{A}$ and polarizability $\mathbf{P}$, as $-\partial_t\mathbf{A}\times\mathbf{P}$, combine into a single partial time-derivative term, $\tau_1 = \partial_t(\mathbf{P}\times\mathbf{A})$, while the interaction between polarizability $\mathbf{P}$ and electric field $\mathbf{E}$ gives $\tau_2 = \mathbf{P}\times\mathbf{E}$. The time average of τ_1 vanishes, resulting zero net transfer spin angular momentum to or from the NS, consequently, the evolution of the spin angular momentum is governed by $\mathbf{P}\times\mathbf{E}$. Notably, we find a highly-symmetrical torque distribution (Fig. 2a), where the incident electromagnetic field breaks the symmetry along x and y axes, and leads to a four-piece distribution that is similar to the scattering pattern of the Stokes parameter S_3.

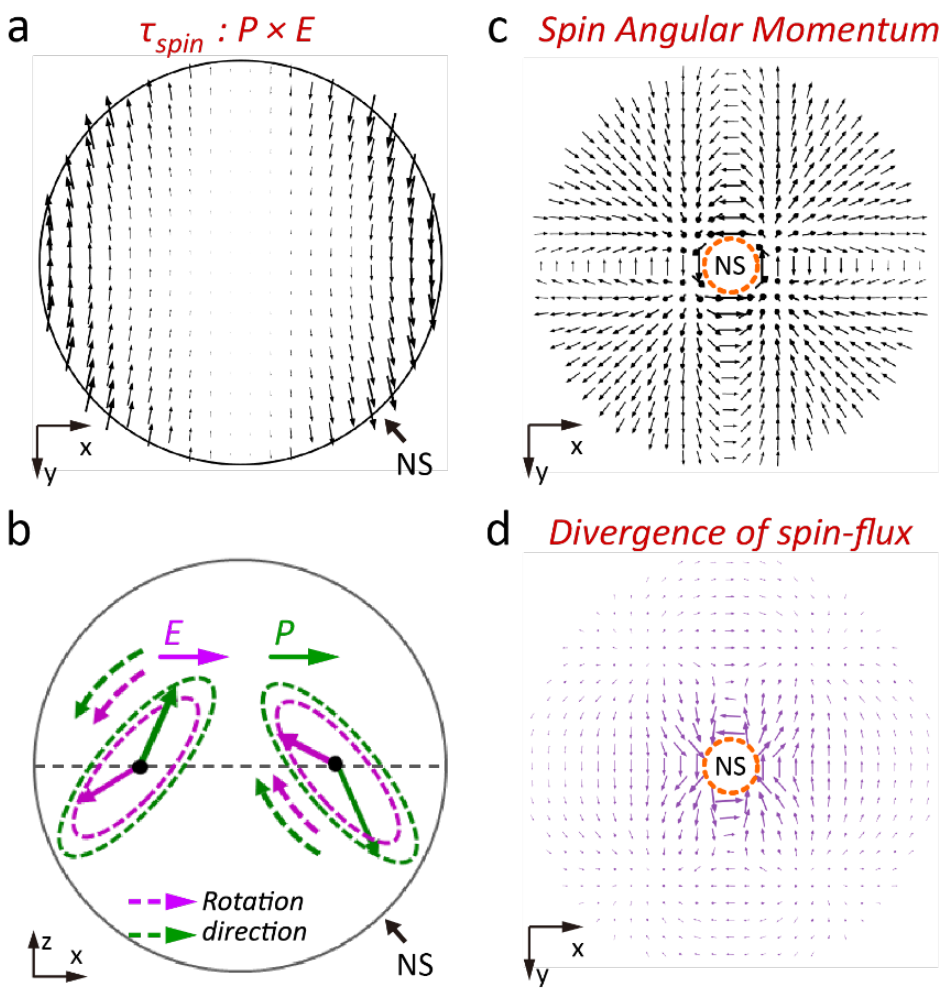

Fig. 2. Computed Spin Related Results According to the SAM Continuity Equation. (a) Time-averaged spin torque inside NS for the interaction between **P** and **E**. (b) Electric field (purple) and induced polarizability (green) inside NS. (c) Calculated SAM density in scattering field outside NS (log scale from 0.1 to 30 $\mathrm{kg \cdot m^{-1} s^{-1}}$). (d) Calculated divergence of spin-flux in scattering field outside NS (log scale from 1.6×10^3 to 1.3×10^6 $\mathrm{kg^2 ms^{-6} A^{-2}}$).

The generated torque drives the polarizability in an elliptical trajectory, which emit light of specific spin in specific directions, and causes the NS depolarization. The spin torque τ^{spin} along $+y$ and $-y$ directions leads to both electric field **E** and polarizability **P** rotating in the xz plane anticlockwise and clockwise, respectively (Figs. 2c,d, S15, Movie S7). We note that the spinning polarizability **P** operates like a circular dipole that prefers to emit LCP light into the direction of τ^{spin}, and RCP into the opposite direction of τ^{spin} (see Fig. S16 and Ref.[36]), which provides an intuitive connection between the four-piece scattering pattern of S_3 and the time-averaged torque distribution in Fig. 2(a). These results coincide well with the SAM and orbital momentum density $\mathbf{p}^o$ distributions(Since the spin momentum density $\mathbf{p}^s$ does not contribute to the energy transport, we replace the Poynting vector **S** with $\mathbf{p}^o$ to understand photon trajectory[Pan 2016]) of the scattering field outside the NS: There is an outward spin angular momentum in the first and third quadrants, while an inward spin angular momentum in the second and fourth quadrants (Fig. 2c); in the scattered OAM, LCP has larger $\mathbf{p}^o$ in the first and third quadrants, while smaller $\mathbf{p}^o$ in the second and fourth quadrants (Fig. 3c).[删去了LCP和RCP分别为outward和inward的说明，在当前情景中读者应该容易分辨．同时删去了"These features explain the origin of the photonic SHE from a NS.", 可能改成类似"These torque driven processes provide a microscopic and dynamic view of SHE of NS scattering"会更好]

[Fig.3 (a)xy平面轨道力 (b)xz平面轨道力 (c)LCP, RCP轨道动量分解 (d)xy平面转换项 (e)45度$\tau^{spin,obrit}$投影(待定)]

STEP1: [Durach 2017: may help]To better understand the elliptical trajectory of

STEP1(alter): To better understand the interaction between light and NS,

STEP2: SOI